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METHOD OF FORMING SEMICONDUCTOR STRUCTURE

Abstract

A method of forming a semiconductor structure includes forming a dielectric stack over a substrate, in which forming the dielectric stack includes forming a first support layer, a first sacrificial layer, a second support layer, a second sacrificial layer and a third support layer in sequence. A first hard mask layer is formed over the dielectric stack. A second hard mask layer is formed over the first hard mask layer. A patterned mask is formed over the second hard mask layer. The first and second hard mask layers are etched using the patterned mask as an etch mask to form first and second hard masks, in which the first hard mask layer is etched faster than the second hard mask layer. An opening is formed in the dielectric stack to expose the substrate. A bottom electrode layer is formed in the opening of the dielectric stack.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a Continuation application of the U.S. application Ser. No. 17/933,121 filed Sep. 18, 2022, which is herein incorporated by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a method of forming a semiconductor structure.

Description of Related Art

[0003] Capacitors are used in a wide variety of semiconductor circuits. For example, the capacitors are used in, for example, DRAM (dynamic random access memory) memory circuits or any other type of memory circuit. DRAM memory circuits are manufactured by replicating millions of identical circuit elements, known as DRAM cells, on a single semiconductor wafer. A DRAM cell is an addressable location that can store one bit (binary digit) of data. In its most common form, a DRAM cell consists of two circuit components: a storage capacitor and an access field effect transistor.

SUMMARY

[0004] One aspect of the present disclosure is a method of forming a semiconductor structure.

[0005] According to some embodiments of the present disclosure, a method of forming a semiconductor structure includes forming a dielectric stack over a substrate, in which forming the dielectric stack includes forming a first support layer, a first sacrificial layer, a second support layer, a second sacrificial layer and a third support layer in sequence. A first hard mask layer is formed over the dielectric stack. A second hard mask layer is formed over the first hard mask layer. A patterned mask is formed over the second hard mask layer. The first hard mask layer and the second hard mask layer are etched using the patterned mask as an etch mask to form a first hard mask and a second hard mask, in which the first hard mask layer is etched faster than the second hard mask layer. An opening is formed in the dielectric stack to expose the substrate. A bottom electrode layer is formed in the opening of the dielectric stack.

[0006] In some embodiments, the method further includes etching the third support layer of the dielectric stack after etching the first hard mask layer and the second hard mask layer.

[0007] In some embodiments, wherein etching the third support layer is performed such that the second sacrificial layer of the dielectric stack is exposed.

[0008] In some embodiments, etching the first hard mask layer and the second hard mask layer is performed such that the first hard mask has a first exposed sidewall and the second hard mask has a second exposed sidewall, and a slope of the second exposed sidewall is substantially the same as a slope of the first exposed sidewall.

[0009] In some embodiments, the first hard mask layer and the second hard mask layer include semiconductor materials.

[0010] In some embodiments, the first hard mask layer includes first dopants having a first conductivity type, and the second hard mask layer includes second dopants having a second conductivity type different from the first conductivity type.

[0011] In some embodiments, the first hard mask layer has a first thickness, the second hard mask layer has a second thickness, and the second thickness is greater than the first thickness.

[0012] In some embodiments, the method further includes removing the patterned mask prior to forming the opening.

[0013] In some embodiments, the method further includes removing the second hard mask prior to

forming the opening.
[0014] In some embodiments, the method further includes forming a high-k dielectric layer along a sidewall of the bottom electrode layer. A top electrode layer along a sidewall of the high-k dielectric layer to define a capacitor including the bottom electrode layer, the high-k dielectric layer and the top electrode layer in the dielectric stack.
[0015] Another aspect of the present disclosure is a method of forming a semiconductor structure.
[0016] According to some embodiments of the present disclosure, a method of forming a semiconductor structure includes forming a dielectric stack over a substrate. A first hard mask and a second hard mask is formed over the dielectric stack, in which the first hard mask includes first dopants having a first conductivity type, and the second hard mask includes second dopants having a second conductivity type different from the first conductivity type. An opening is formed in the dielectric stack using the first hard mask as an etch mask to expose the substrate. A bottom electrode layer is formed in the opening of the dielectric stack.
[0017] In some embodiments, forming the first hard mask and the second hard mask is performed such that the first hard mask is in contact with the dielectric stack and the second hard mask is in contact with the first hard mask.
[0018] In some embodiments, the first hard mask and the second hard mask include polysilicon materials.
[0019] In some embodiments, a dopant concentration of the first dopants is in a range of 0.5% to 1.5%, and a dopant concentration of the second dopants is in a range of 0.5% to 1.5%.
[0020] In some embodiments, the first hard mask has a first thickness, the second hard mask has a second thickness, and the second thickness is greater than the first thickness.
[0021] In some embodiments, the method further includes removing the first hard mask prior to forming the opening in the dielectric stack.
[0022] In some embodiments, forming the dielectric stack includes forming a first sacrificial layer over the substrate. A second sacrificial layer is formed over the first sacrificial layer.
[0023] In some embodiments, forming the dielectric stack further includes forming a first support layer over the substrate. A second support layer is formed over the first sacrificial layer. A third support layer is formed over the second sacrificial layer.
[0024] In some embodiments, the method further includes removing the second sacrificial layer after forming the bottom electrode layer. The first sacrificial layer is removed to expose the first support layer.
[0025] In some embodiments, the method further includes forming a high-k dielectric layer along a sidewall of the bottom electrode layer. A top electrode layer along a sidewall of the high-k dielectric layer to define a capacitor including the bottom electrode layer, the high-k dielectric layer and the top electrode layer in the dielectric stack.
[0026] In the aforementioned embodiments, since the first hard mask layer is etched faster than the second hard mask layer, a profile of the first hard mask can be improved. As such, a profile of the capacitor formed in the subsequent etching and deposition processes can be improved and decreased capacitance of the capacitor can be avoided.
[0027] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0028] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the

standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
[0029] FIGS. **1-11** illustrate cross-section views of intermediate stages of a process of a semiconductor structure in accordance with some embodiments of the present disclosure.
DETAILED DESCRIPTION
[0030] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.
[0031] As used herein, “around,” “about,” “approximately,” or “substantially” shall generally mean within 20 percent, or within 10 percent, or within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated.
[0032] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0033] FIGS. **1-11** illustrate cross-section views of intermediate stages of a process of a semiconductor structure in accordance with some embodiments of the present disclosure. Referring to FIG. **1**, a dielectric stack DS is formed over a substrate **110**. In greater details, forming the dielectric stack DS over the substrate **110** includes forming a first support layer **120** over the substrate **110**, forming a first sacrificial layer **130** over the first support layer **120**, forming a second support layer **140** over the first sacrificial layer **130**, forming a second sacrificial layer **150** over the second support layer **140**, and forming a third support layer **160** over the second sacrificial layer **150**. In other words, the dielectric stack DS includes the first support layer **120**, the first sacrificial layer **130**, the second support layer **140**, the second sacrificial layer **150** and the third support layer **160** formed in sequence.
[0034] The substrate **110** includes a device region DR and a peripheral region PR. The device region DR is a region where a semiconductor structure (e.g., capacitors) is formed. The peripheral region PR is adjacent to the device region DR. For example, the peripheral region PR surrounds the device region DR. The substrate **110** includes a dielectric layer **112**. In some embodiments, the dielectric layer **112** is an interlayer dielectric (ILD) layer or an intermetal dielectric (IMD) layer. The dielectric layer **112** may be a low-k dielectric layer made of a low dielectric constant (k) material, a very low-k material, or a combination thereof. In some embodiments, the dielectric layer **112** includes nitride (e.g., silicon nitride), or other suitable dielectric material. The substrate **110** further includes landing pads **114** and bit line contacts **116** formed in the dielectric layer **112**, in which the landing pads **114** and the bit line contacts **116** are located in the device region DR of the substrate **110**, and the landing pads **114** and the bit line contacts **116** are not located in the peripheral region PR of the substrate **110**. The landing pads **114** and/or the bit line contacts **116** may include aluminum, aluminum alloys, copper, copper alloys, titanium, titanium nitride, tantalum, tantalum nitride, tungsten, cobalt, the like, or combinations thereof. In some embodiments, the substrate **110** further includes transistors or other similar components. As a result, capacitors (e.g., capacitors Ca in FIG. **11**) subsequently formed in the dielectric stack DS are connected to the other components (e.g., transistors) in the substrate **110** through the landing pads **114**.
[0035] In some embodiments, the first support layer **120**, the second support layer **140** and the third support layer **160** include nitride, such as silicon nitride. In some embodiments, the first sacrificial layer **130** and the second sacrificial layer **150** include oxide. The first sacrificial layer **130** and the

second sacrificial layer **150** may be made of different materials. When forming the first sacrificial layer **130**, dopants are doped in the first sacrificial layer **130**, and the dopants include boron, phosphorus, or combinations thereof. For example, the first sacrificial layer **130** is made of boro-phospho-silicate-glass (BPSG) which is silicon oxide doped with boron and phosphorous. In some embodiments, the second sacrificial layer **150** is made of tetraethoxysilane (TEOS) oxide, or other suitable oxide material.

[0036] Referring to FIG. **2**, after the dielectric stack DS is formed over the substrate **110**, a first hard mask layer **170′** is formed over the dielectric stack DS. The first hard mask layer **170′** is in contact with the third support layer **160** of the dielectric stack DS. The third support layer **160** is located between the first hard mask layer **170′** and the second sacrificial layer **150**. A second hard mask layer **180′** is then formed over the first hard mask layer **170′**. The second hard mask layer **180′** is in contact with the first hard mask layer **170′**. The first hard mask layer **170′** is located between the second hard mask layer **180′** and the third support layer **160**. Thereafter, a patterned mask **190** is formed over the second hard mask layer **180′**. In some embodiments, the formation of the patterned mask **190** includes forming a mask layer over the second hard mask layer **180′**, and then patterning the mask layer (e.g., performing a dry etching process) to expose a portion of the second hard mask layer **180′**. In some embodiments, the patterned mask **190** includes a material different from the first hard mask layer **170′** or the second hard mask layer **180′**. For example, the patterned mask **190** includes oxide (e.g., tetraethoxysilane (TEOS) oxide), the first hard mask layer **170′** includes semiconductor materials (e.g., polysilicon), and the second hard mask layer **180′** includes semiconductor materials (e.g., polysilicon).

[0037] Referring to FIG. **2** and FIG. **3**, performing an etching process on the first hard mask layer **170′** and the second hard mask layer **180′** to form a first hard mask **170** and a second hard mask **180**, in which the first hard mask layer **170′** is etched faster than the second hard mask layer **180′**. As such, a profile of the first hard mask **170** can be improved. Further, a profile of the capacitor (e.g., the capacitor Ca in FIG. **11**) formed in the subsequent etching and deposition processes can be improved and decreased capacitance of the capacitor can be avoided. The improved profile of the capacitor may enhance the balance of the capacitor and avoid the problem of pattern abnormal.

[0038] In some embodiments, the etching process is performed to etch a portion of the second hard mask layer **180′** and a portion of the first hard mask layer **170′** using the patterned mask **190** as an etch mask. Then, performing an etching process to etch a portion of the third support layer **160** of the dielectric stack DS. As such, openings O**1** are formed in the second hard mask **180**, the first hard mask **170** and the third support layer **160**. Since the first hard mask layer **170′** is etched faster than the second hard mask layer **180′**, the first hard mask **170** is laterally expanded, thereby improving the profile of the first hard mask **170**. In some embodiments, each of the openings O**1** has a tapered profile (e.g., U-shaped profile). Each of the openings O**1** may have a first maximum width in the third support layer **160**, a second maximum width in the first hard mask **170** and a third maximum width in the second hard mask **180**, in which the first maximum width is smaller than the second maximum width, and the second maximum width is substantially equal to the third maximum width. In other words, the second hard mask **180** has a second exposed sidewall **181** facing the openings O**1** and the first hard mask **170** has a first exposed sidewall **171** facing the openings O**1**, in which a slope of the second exposed sidewall **181** is substantially the same as a slope of the first exposed sidewall **171**. For example, the substrate **110** has a substantially planar bottom surface that extends along a first direction, the second exposed sidewall **181** extends along a second direction perpendicular to the first direction, and the first exposed sidewall **171** extends along the second direction. In some embodiments, the openings O**1** further exposes the second sacrificial layer **150** of the dielectric stack DS.

[0039] In some embodiments, the first hard mask layer **170′** (first hard mask **170**) has a first thickness T**1** and the second hard mask layer **180′** (second hard mask **180**) has a second thickness T**2**, in which the second thickness T**2** is greater than the first thickness T**1**. For example, the first thickness T**1** of the first hard mask layer **170′** is in a range of about 80 nanometers to about 40 nanometers (e.g., 60 nanometers), and the second thickness T**2** of the second hard mask layer **180′** is in a range of about 100 nanometers to about 140 nanometers (e.g., 120 nanometers). In some embodiments, a ratio of the second thickness T**2** to the first thickness T**1** is in a range of about 1.5 to about 2.5 (e.g., 2). If the ratio of the second thickness T**2** to the first thickness T**1** is out of the above selected ranges, the profile of the first hard mask **170** would not formed uniformly. For example, openings (e.g., openings O**1**) in the first hard mask **170** would be not formed through the first hard mask **170** and would not laterally expanded, and thus openings for capacitors that is formed in the subsequent etching processes would be too smaller, thereby causing the imbalance of the capacitors that is formed in the subsequent deposition processes.

[0040] The first hard mask layer **170′** and the second hard mask layer **180′** may include semiconductor materials, such as polysilicon or other suitable materials. The first hard mask layer **170′** may include first dopants having a first conductivity type (e.g., N-type in this case) such as phosphorous (P), arsenic (As), antimony (Sb), combinations thereof, or the like. In some embodiments, an implantation process is performed on the first hard mask layer **170′**, followed by an annealing process to activate the implanted first dopants of the first hard mask layer **170′**. The second hard mask layer **180′** may include second dopants having second conductivity type (e.g., P-type in this case) such as boron (B), BF.sub.2, BF.sub.3, combinations thereof, or the like. In some embodiments, an implantation process is performed on the second hard mask layer **180′**, followed by an annealing process to activate the implanted second dopants of the second hard mask layer **180′**. In some embodiments, the second dopants of the second hard mask layer **180′** have different conductivity type from the first dopants of the first hard mask layer **170′**.

[0041] Since the second conductivity type of the second dopants of the second hard mask layer **180′** is different from the first conductivity type of the first dopants of the first hard mask layer **170′**, the second hard mask layer **180′** and the first hard mask layer **170′** are etched with different etching rate. In greater details, since the first hard mask layer **170′** includes the first dopants (e.g., N-type in this case) and the second hard mask layer **180′** includes the second dopants (e.g., P-type in this case), a hardness (or density) of the second hard mask layer **180′** is greater than a hardness (or density) of the first hard mask layer **170′**. Further, since the outer electrons of the first dopants (e.g., N-type) of the first hard mask layer **170′** are more likely or easily to react with dry etchant (e.g., Br ion) because of their electronegativity than that of the second dopants (e.g., P-type) of the second hard mask layer **180′**, the first hard mask layer **170′** with the first dopants (e.g., N-type) is etched faster than the second hard mask layer **180′** with the second dopants (e.g., P-type). As a result, the profile of the first hard mask **170** can be improved. If the first hard mask layer **170′** is not etched faster (e.g., lower or at the same etching rate) than the second hard mask layer **180′**, the openings O**1** would have sharp profiles (e.g., V-shaped profile) such that the capacitors formed in the subsequent processes would not aligned with each other (e.g., a length of each capacitors is not identical). In some embodiments, a dopant concentration of the first dopants of the first hard mask layer **170′** is in a range of about 0.5% to about 1.5%, and a dopant concentration of the second dopants of the second hard mask layer **180′** is in a range of about 0.5% to about 1.5%. If the dopant concentration of the first dopants and/or the dopant concentration of second dopants are out of the above selected ranges, the etching rate of the first hard mask layer **170′** and the etching rate of the second hard mask layer **180′** would be difficult to control (e.g., difference between the etching rate of the first hard mask layer **170′** and the etching rate of the second hard mask layer **180′** is not significant enough), and thus the improved profile of the first hard mask **170** would not achieved.

[0042] In some embodiments, etching the first hard mask layer **170′** and the second hard mask layer **180′** are performed using the patterned mask **190** as an etch mask by a dry etching process. For example, a dry etchant for the dry etching process includes HBr, NH.sub.3, O.sub.2, combinations thereof, or other suitable gases. The second hard mask layer **180′** and the first hard mask layer **170′** are etched using the same dry etchant (e.g., HBr, NH.sub.3, or O.sub.2). In some

embodiments where the dry etchant is HBr, the flow of the dry etching process is in a range of about 150 sccm to about 200 sccm. In some embodiments where the dry etchant is NH.sub.3, the flow of the dry etching process is in a range of about 30 sccm to about 50 sccm. In some embodiments where the dry etchant is O.sub.2, the flow of the dry etching process is in a range of about 20 sccm to about 35 sccm. If the flow of the dry etching process is out of the above selected ranges, the profile of the first hard mask **170** would not be formed uniformly (e.g., the openings O**1** are too large). In some embodiments, the dry etching process utilizes a power in a range of about 1500 Watts to about 200 Watts. If the power of the dry etching process is out of the above selected ranges, the openings O**1** would be under-etch (e.g., the openings O**1** not through the first hard mask layer **170**′). In some embodiments, etching the third support layer **160** is performed by a dry etching process after etching the first hard mask layer **170**′ and the second hard mask layer **180**′. For example, a dry etchant for the dry etching process of the third support layer **160** includes SF.sub.6, Cl.sub.2, combinations thereof, or other suitable gases. The third support layer **160** is etched using different dry etchant from the second hard mask layer **180**′ (or the first hard mask layer **170**′).
[0043] Referring to FIG. **3** and FIG. **4**, after forming the openings O**1**, openings O**2** are formed in the dielectric stack DS to expose the substrate **110**. In other words, the dielectric stack DS is continued to be etched along the openings O**1** to form the openings O**2**. The opening O**2** is formed in the dielectric stack DS using the first hard mask **170** as an etch mask. The openings O**2** may expose sidewalls DS**1** of the dielectric stack DS. In some embodiments, the patterned mask **190** and the second hard mask **180** are removed in sequence prior to forming the openings O**2**.
[0044] In some embodiments, etching the dielectric stack DS to form the openings O**2** includes multiple etching processes. For example, the second sacrificial layer **150** is etched to expose the second support layer **140**, the second support layer **140** is etched to expose the first sacrificial layer **130**, the first sacrificial layer **130** is etched to expose the first support layer **120**, and the first support layer **120** is etched to expose the landing pads **114** of the substrate **110**.
[0045] Referring to FIG. **4** and FIG. **5**, after forming the openings O**2** in the dielectric stack DS, the first hard mask **170** is removed such that a top surface **161** of the third support layer **160** of the dielectric stack DS is exposed. In some embodiments, removing the first hard mask **170** is performed by a dry etching process.
[0046] Referring to FIGS. **6-11**, capacitors Ca are formed in the openings O**2** (see FIG. **5**) of the dielectric stack DS. Discussed in greater details, referring to FIG. **5** and FIG. **6**, bottom electrode layers **200** are formed in the openings O**2** of the dielectric stack DS. The bottom electrode layers **200** may include curved portions **202** and horizontal portions **204** connected to the curved portions **202**, in which the curved portions **202** are located along the sidewall DS**1** of the dielectric stack DS and in contact with the landing pads **114** of the substrate **110**, and the horizontal portions **204** is located over the curved portions **202** and the third support layer **160** of the dielectric stack DS. In some embodiments, the bottom electrode layers **200** include titanium nitride (TiN) or other suitable conductive materials.
[0047] After the bottom electrode layers **200** are formed, a mask **210** is formed over the bottom electrode layers **200**. The mask **210** is in contact with the bottom electrode layers **200**. The formation of the mask **210** may include forming a mask layer over the horizontal portion **204** of bottom electrode layer **200**, and then patterning the mask layer to expose the horizontal portion **204** of bottom electrode layer **200**. The bottom electrode layers **200** are etched to expose underlying third support layer **160** using the mask **210** as an etch mask. In some embodiments, the mask **210** and the third support layer **160** include the same materials, such as silicon nitride.
[0048] Thereafter, a patterned mask **220** is formed over the mask **210**. The formation of the patterned mask **220** may include forming a mask layer over the mask **210**, and then patterning the mask layer to expose a portion of the mask **210**. The patterned mask **220** is in contact with the mask **210**. In some embodiments, the patterned mask **220** includes oxide, such as tetraethoxysilane

(TEOS) oxide, silicon oxide or other suitable materials. In some other embodiments, the patterned mask **220** includes the same material as that of the first sacrificial layer **130** or the second sacrificial layer **150**. The patterned mask **220** may have a different material from that of the mask **210**. For example, the patterned mask **220** includes silicon oxide, and the mask **210** includes silicon nitride.

[0049] Referring to FIG. **6** and FIG. **7**, the mask **210** is etched to expose the bottom electrode layers **200** using the patterned mask **220** as an etch mask. Further, the third support layer **160** is etched to expose the second sacrificial layer **150** using the patterned mask **220** as the etch mask. In some embodiments, the mask **210** and the third support layer **160** are etched simultaneously by using one etching process.

[0050] Thereafter, performing an etching process to remove an entirety of the second sacrificial layer **150** of the dielectric stack DS such that the second support layer **140** is exposed. In some embodiments, the second sacrificial layer **150** is removed by using a wet etching process, and an etch solution thereof includes fluoride-based solution, such as hydrogen fluoride (HF). After removing the second sacrificial layer **150**, spaces S**1** are formed between the second support layer **140** and the third support layer **160**. In some embodiments, a portion of the second support layer **140** is etched to align with the third support layer **160**. In some embodiments, during the etching process, the patterned mask **220** is simultaneously removed to expose the mask **210**. In other words, the patterned mask **220** and the second sacrificial layer **150** are removed simultaneously using one etching process.

[0051] Referring to FIG. **7** and FIG. **8**, the mask **210** is removed after removing the second sacrificial layer **150**. In some embodiments, etching the mask **210** is performed by a dry etching process. Thereafter, the horizontal portions **204** of the bottom electrode layers **200** are removed, while leaving the curved portions **202** of the bottom electrode layers **200** remained. As such, each of the bottom electrode layers **200** has U-shaped profile. The top surface **161** of the third support layer **160** is exposed and sidewalls **201** of the bottom electrode layers **200** are exposed through the openings O**2**. In some embodiments, etching the horizontal portions **204** of the bottom electrode layers **200** is performed by a dry etching process.

[0052] Referring to FIG. **8** and FIG. **9**, an entirety of the first sacrificial layer **130** of the dielectric stack DS is removed. In some embodiments, an etching process is performed to remove the first sacrificial layer **130**. For example, the first sacrificial layer **130** is removed by using a wet etching process, and an etch solution thereof includes fluoride-based solution, such as hydrogen fluoride (HF). After removing the first sacrificial layer **130**, spaces S**2** are formed between the second support layer **140** and the first support layer **120**. The first support layer **120**, the second support layer **140** and the third support layer **160** are connected by the bottom electrode layers **200**.

[0053] Referring to FIG. **10**, high-k dielectric layers **230** are formed along sidewalls of the bottom electrode layers **200**. The high-k dielectric layers **230** may be formed along the sidewalls **203** of the bottom electrode layers **200** in the spaces S**1** and the spaces S**2**. The high-k dielectric layers **230** may also be formed along the sidewalls **201** of the bottom electrode layers **200** in the openings O**2** of the second dielectric stack DS. Further, the high-k dielectric layers **230** may be formed along the top surface and the bottom surface of the second support layer **140** and the third support layer **160**, and the high-k dielectric layers **230** may be formed along the top surface of the first support layer **120**. In some embodiments, the high-k dielectric layers **230** include hafnium oxide (HfO). In various examples, the high-k dielectric layers **230** include metal oxide (such as HfSiO.sub.2, ZnO, ZrO.sub.2, Ta.sub.2O.sub.5, Al.sub.2O.sub.3, or the like), metal nitride, or combinations thereof.

[0054] Referring to FIG. **10** and FIG. **11**, first electrode layers **240** are formed along the high-k dielectric layers **230**. The first electrode layers **240** may be formed along sidewalls **233** of the high-k dielectric layers **230** in the spaces S**1** and the spaces S**2**. The first electrode layers **240** may also be formed along sidewalls **231** of the high-k dielectric layers **230** in the openings O**2** of the dielectric stack DS. In some embodiments, the first electrode layers **240** are located over a horizontal surface of the high-k dielectric layers **230**. The first electrode layers **240** may include titanium nitride (TiN), or other suitable conductive materials. In some embodiments, the first electrode layers **240** and the bottom electrode layers **200** include the same materials.

[0055] After the first electrode layers **240** are formed, semiconductor layers **250** are formed in the spaces S**1** between the first support layer **120** and the second support layer **140**, and the spaces S**2** between the second support layer **140** and the third support layer **160**. The semiconductor layers **250** may entirely fill the openings O**2** of the dielectric stack DS. The semiconductor layers **250** may also be in contact and cover the first electrode layers **240** over the third support layer **160**. The semiconductor layers **250** may include polysilicon or other suitable semiconductive materials. The semiconductor layers **250** have a material different from that of the first electrode layers **240** or bottom electrode layers **200**.

[0056] After forming the semiconductor layers **250** are formed, second electrode layers **260** are formed over the semiconductor layers **250**. The second electrode layers **260** are formed along a top surface and sidewalls of the semiconductor layers **250**. The second electrode layers **260** may also be in contact and cover the semiconductor layers **250**. The second electrode layers **260** may include metal (e.g., tungsten) or other suitable conductive materials. The second electrode layers **260** have a material different from that of the first electrode layers **240** or bottom electrode layers **200**. The first electrode layers **240**, the semiconductor layers **250** and the second electrode layers **260** may have the same electrical potential. In some embodiments, the first electrode layers **240** are referred as top electrode layers of capacitors Ca. In some other embodiments, the first electrode layers **240** and the semiconductor layers **250** are referred as the top electrode layers of the capacitors Ca. Alternatively, the first electrode layers **240**, the semiconductor layers **250** and the second electrode layers **260** are referred as the top electrode layers of the capacitors Ca. After the top electrode layers (i.e., the first electrode layers **240**, the semiconductor layers **250** and/or the second electrode layers **260**) are formed, the capacitors Ca including the bottom electrode layers **200**, the high-k dielectric layers **230** and the top electrode layers are defined in the openings O**2** of the dielectric stack DS and in the spaces S**1** and the spaces S**2**.

[0057] In some embodiments, a semiconductor structure includes a plurality of support layers (i.e., the first support layer **120**, the second support layer **140** and the third support layer **160**) and the capacitors Ca. The first support layer **120**, the second support layer **140** and the third support layer **160** are arranged from bottom to top, and the first support layer **120**, the second support layer **140** and the third support layer **160** are spaced apart from each other. In other words, the third support layer **160** is located above the second support layer **140**, and the second support layer **140** is located above the first support layer **120**. In some embodiments, a width of the third support layer **160** is greater than a width of the second support layer **140** or a width of the first support layer **120**. Each of the capacitors Ca includes the bottom electrode layer **200**, the high-k dielectric layer **230** and the top electrode layer (i.e., the first electrode layer **240**, the semiconductor layer **250** and/or the second electrode layer **260**). The capacitors Ca may have U-shaped profile in the dielectric stack DS. It is noted that the capacitors Ca in FIG. **11** are illustrative, and the capacitors Ca are not limited to the configuration illustrated in FIG. **11**. For example, additional layers are included in the capacitors Ca.

[0058] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0059] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A method of forming a semiconductor structure, comprising: forming a dielectric stack over a substrate, wherein forming the dielectric stack comprises forming a first support layer, a first sacrificial layer, a second support layer, a second sacrificial layer and a third support layer in sequence; forming a first hard mask layer over the dielectric stack; forming a second hard mask layer over the first hard mask layer, wherein the first hard mask layer has a first thickness, the second hard mask layer has a second thickness, and the second thickness is greater than the first thickness; forming a patterned mask over the second hard mask layer; etching the first hard mask layer and the second hard mask layer using the patterned mask as an etch mask to form a first hard mask and a second hard mask, wherein the first hard mask layer is etched at a different rate than the second hard mask layer; forming an opening in the dielectric stack to expose the substrate; and forming a bottom electrode layer in the opening of the dielectric stack.
2. The method of claim 1, further comprising: etching the third support layer of the dielectric stack after etching the first hard mask layer and the second hard mask layer.
3. The method of claim 2, wherein etching the third support layer is performed such that the second sacrificial layer of the dielectric stack is exposed.
4. The method of claim 1, wherein etching the first hard mask layer and the second hard mask layer is performed such that the first hard mask has a first exposed sidewall and the second hard mask has a second exposed sidewall, and a slope of the second exposed sidewall is substantially the same as a slope of the first exposed sidewall.
5. The method of claim 1, wherein the first hard mask layer and the second hard mask layer comprise semiconductor materials.
6. The method of claim 1, wherein the first hard mask layer comprises first dopants having a first conductivity type, and the second hard mask layer comprises second dopants having a second conductivity type different from the first conductivity type.
7. The method of claim 1, wherein the first hard mask layer is etched faster than the second hard mask layer.
8. The method of claim 1, further comprising: removing the patterned mask prior to forming the opening.
9. The method of claim 1, further comprising: removing the second hard mask prior to forming the opening.
10. The method of claim 1, further comprises: forming a high-k dielectric layer along a sidewall of the bottom electrode layer; and forming a top electrode layer along a sidewall of the high-k dielectric layer to define a capacitor comprising the bottom electrode layer, the high-k dielectric layer and the top electrode layer in the dielectric stack.
11. A method of forming a semiconductor structure, comprising: forming a dielectric stack over a substrate; forming a first hard mask and a second hard mask over the dielectric stack, wherein the first hard mask has a first thickness, the second hard mask has a second thickness, and the second thickness is greater than the first thickness; forming an opening in the dielectric stack using the first hard mask as an etch mask to expose the substrate; and forming a bottom electrode layer in the opening of the dielectric stack.
12. The method of claim 11, wherein forming the first hard mask and the second hard mask is performed such that the first hard mask is in contact with the dielectric stack and the second hard mask is in contact with the first hard mask.
13. The method of claim 11, wherein the first hard mask and the second hard mask comprise polysilicon materials.
14. The method of claim 11, wherein the first hard mask comprises first dopants having a first conductivity type, and the second hard mask comprises second dopants having a second

conductivity type different from the first conductivity type.

15. The method of claim 14, wherein a dopant concentration of the first dopants is in a range of 0.5% to 1.5%, and a dopant concentration of the second dopants is in a range of 0.5% to 1.5%.

16. The method of claim 11, further comprising: removing the first hard mask prior to forming the opening in the dielectric stack.

17. The method of claim 11, wherein forming the dielectric stack comprises: forming a first sacrificial layer over the substrate; and forming a second sacrificial layer over the first sacrificial layer.

18. The method of claim 17, wherein forming the dielectric stack further comprises: forming a first support layer over the substrate; forming a second support layer over the first sacrificial layer; and forming a third support layer over the second sacrificial layer.

19. The method of claim 18, further comprising: removing the second sacrificial layer after forming the bottom electrode layer; and removing the first sacrificial layer to expose the first support layer.

20. The method of claim 11, further comprises: forming a high-k dielectric layer along a sidewall of the bottom electrode layer; and forming a top electrode layer along a sidewall of the high-k dielectric layer to define a capacitor comprising the bottom electrode layer, the high-k dielectric layer and the top electrode layer in the dielectric stack.